Difference in Differences

Joan Llull

I. Introduction

With data from a randomized experiment, the simple comparison of the mean outcome in treatment and control groups (a "difference" estimator) provides an unbiased and consistent estimate of the average treatment effect (ATE), as discussed in Chapter 2. Randomization ensures treated and control groups are comparable: there are no systematic differences in the underlying determinants of the outcome in the absence of treatment (Y_{0i}), and, hence, no selection bias.

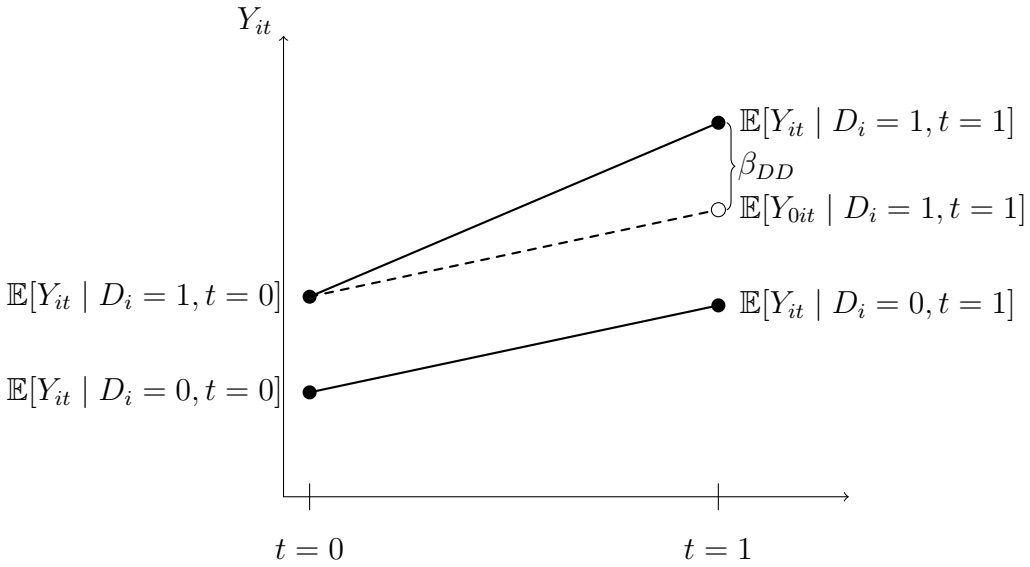
In many relevant applications, treatment assignment is not random. In later chapters we dealt with departures from independence through alternative strategies: in Chapter 3 (selection on observables) and in sharp RD designs in Chapter 5, we proposed methods that attempt to construct comparable treated and control groups using observed covariates or local comparisons around a cutoff. In Chapter 4 and in fuzzy RD designs in Chapter 5, we used instrumental variables to identify causal effects in the presence of endogeneity. However, good instruments are hard to find, and it is often desirable to use other sources of variation.

The approach in this chapter uses *changes over time* to difference out *time-invariant* observed and unobserved confounders. The key identifying idea is that, even if treated and control groups differ in levels (selection bias), they may share the same counterfactual *trend* in the absence of treatment. Under this *parallel trends* (or *common trends*) assumption, the outcome evolution in the control group can be used to construct the counterfactual outcome evolution for the treated group had treatment not occurred. The resulting estimand is a *difference-in-differences* (DiD): the change in outcomes for treated units minus the change in outcomes for controls.

II. Difference in Differences Setup

Consider a two-period setup $t \in \{0, 1\}$ and two groups indexed by $D_i \in \{0, 1\}$, where $D_i = 1$ denotes membership in the treated group and $D_i = 0$ membership in the control group. Let Y_{it} be the observed outcome and let Y_{0it} and Y_{1it} denote the potential outcomes without and with treatment, respectively. Treatment occurs

Figure 1. Difference-in-differences intuition in the two-group, two-period case.



only in period $t = 1$ and only for the treated group, so we observe:

$$Y_{it} = \begin{cases} Y_{0it}, & \text{if } t = 0 \text{ (both groups untreated)}, \\ Y_{1it}, & \text{if } t = 1 \text{ and } D_i = 1, \\ Y_{0it}, & \text{if } t = 1 \text{ and } D_i = 0. \end{cases} \tag{1}$$

Thus, the missing counterfactual is the untreated potential outcome for treated units in period $t = 1$, namely $\mathbb{E}[Y_{0i1} \mid D_i = 1]$.

Figure 1 illustrates the basic two-group, two-period difference-in-differences setup. The vertical axis represents the outcome Y_{it} and the horizontal axis time. The lower line corresponds to the control group, while the upper line corresponds to the treated group. In period $t = 0$, neither group is treated, and we observe a baseline level difference between them. Between $t = 0$ and $t = 1$, the control group experiences a change that reflects the counterfactual time trend in the absence of treatment. The dashed line extending the treated group's pre-treatment trajectory using the control group's slope represents the counterfactual outcome $\mathbb{E}[Y_{0i1} \mid D_i = 1]$: what the treated group would have experienced at $t = 1$ absent treatment under the parallel trends assumption. The vertical distance between the observed treated outcome at $t = 1$ and this counterfactual is the difference-in-differences estimand β_{DD}. Graphically, DiD subtracts the baseline gap (level difference at $t = 0$) and the common time trend (captured by the control group) from the observed post-treatment difference, isolating the average treatment effect on the treated.

The fundamental DiD identifying assumption is the *parallel trends* assumption:

$$\mathbb{E}[Y_{0i1} - Y_{0i0} \mid D_i = 1] = \mathbb{E}\left[Y_{0i1} - Y_{0i0} \mid D_i = 0\right]. \tag{2}$$

That is, in the absence of treatment, the treated and control groups would have experienced the same average change in outcomes from $t = 0$ to $t = 1$.

Under (2), we can predict the counterfactual untreated outcome for treated units at $t = 1$ using the control-group trend:

$$\mathbb{E}[Y_{0i1} \mid D_i = 1] = \underbrace{\mathbb{E}[Y_{i1} \mid D_i = 0]}_{\text{controls at } t=1} + \underbrace{\{\mathbb{E}[Y_{i0} \mid D_i = 1] - \mathbb{E}[Y_{i0} \mid D_i = 0]\}}_{\text{baseline gap at } t=0}. \tag{3}$$

The DiD estimand is then:

$$\beta_{DD} \equiv \{\mathbb{E}[Y_{i1} \mid D_i = 1] - \mathbb{E}[Y_{i0} \mid D_i = 1]\} - \{\mathbb{E}[Y_{i1} \mid D_i = 0] - \mathbb{E}[Y_{i0} \mid D_i = 0]\}. \tag{4}$$

This is the difference between the change in average outcomes for the treated group and the change for the control group. Under parallel trends, β_{DD} identifies the average treatment effect on the treated (ATT):

$$\beta_{DD} = \mathbb{E}[Y_{1i1} - Y_{0i1} \mid D_i = 1]. \tag{5}$$

In words, DiD recovers the average causal effect of treatment on those units that are treated.

Difference-in-differences is conceptually simple, but its credibility relies on assumptions that can fail in practice. Key threats to identification include:

- *Parallel trends violations.* Treated and control units would have evolved differently even in the absence of treatment. Importantly, this can occur *even if pre-trends look parallel* over a limited pre-period window.

- *Anticipation and dynamic responses.* Units may adjust behavior before treatment begins (anticipation), or effects may evolve over time (dynamic treatment effects). Both phenomena complicate the interpretation of a single "post" effect.

- *Composition changes and SUTVA violations.* Migration, spillovers, attrition, or changes in who is observed over time can invalidate comparisons. For repeated cross-sections, changes in sample composition over time are an additional concern.

- *Time-varying confounders correlated with treatment timing.* Policies may be bundled with other shocks; if those shocks affect treated units differentially in the post period, parallel trends fails.

These threats motivate the use of placebo checks, event-study diagnostics, richer control strategies, and alternative estimators (synthetic control, modern DiD methods) when staggered adoption and heterogeneous effects are present.

III. Difference in Differences in the Regression Context

The DiD estimand (4) can be obtained as the coefficient on the interaction term in the regression:

$$Y_{it} = \beta_0 + \beta_D D_i + \beta_T T_t + \beta_{DD}(D_i \times T_t) + U_{it}, \tag{6}$$

where $D_i = 1$ for treated-group units and $T_t = 1$ in the post-treatment period ($t = 1$), with $T_t = 0$ otherwise. As in previous chapters, it is useful to interpret each coefficient as a group-time mean: β_0 is $\mathbb{E}[Y_{it}|D_i = 0, t = 0]$, $\beta_0 + \beta_D = \mathbb{E}[Y_{it}|D_i = 1, t = 0]$, $\beta_0 + \beta_T = \mathbb{E}[Y_{it}|D_i = 0, t = 1]$, and β_{DD} is the difference in differences coefficient (4).

Regression (6) can be expanded in several ways:

- *Multiple periods.* With more time periods, it is standard to include a full set of time fixed effects δ_t. Regarding treatment, one may (i) interact treatment status with a single post-treatment indicator (a single average post effect), or (ii) interact treatment with a full set of event-time dummies to obtain dynamic effects (see Section VII).

- *Controls.* Adding covariates X_{it} can improve precision and absorb some time-varying observables, but it does *not* replace the parallel trends assumption. Controls are most credible when they are plausibly exogenous and capture differential composition changes.

- *Repeated cross-sections vs. panel data.* Panel data are not required for DiD; repeated cross-sections can suffice. However, with repeated cross-sections the researcher must maintain that sample composition is comparable over time (or must explicitly model composition changes). Panel data avoid this issue by construction, but introduce their own concerns (attrition).

- *Placebo exercises.* A placebo analysis simulates the DiD design at a time or for a group that resembles the treated setting but was not actually treated

(e.g. "fake" treatment dates or groups). A strong placebo failure is evidence against the identifying assumptions.

IV. Triple Differences Model

Sometimes the standard DiD assumption may fail because, even absent treatment, treated and control groups would have different trends. A *triple-differences* (DDD or 3D) strategy can help when there exists an additional dimension along which a second control comparison is available.

Let G_i denote group membership (e.g., parents vs. non-parents, young vs. old, or another sociodemographic characteristic). A standard DDD regression is:

$$\begin{aligned} Y_{it} = \beta_0 + \beta_D D_i + \beta_T T_t + \beta_G G_i + \beta_{GD}(G_i \times D_i) \\ + \beta_{GT}(G_i \times T_t) + \beta_{DT}(D_i \times T_t) + \beta_{3D}(G_i \times D_i \times T_t) + U_{it}. \end{aligned} \tag{7}$$

The coefficient β_{3D} is the triple-differences estimand: it differences out time effects, treated-group level differences, and group-specific level differences, while also allowing for group-specific time shocks captured by $G_i \times T_t$.

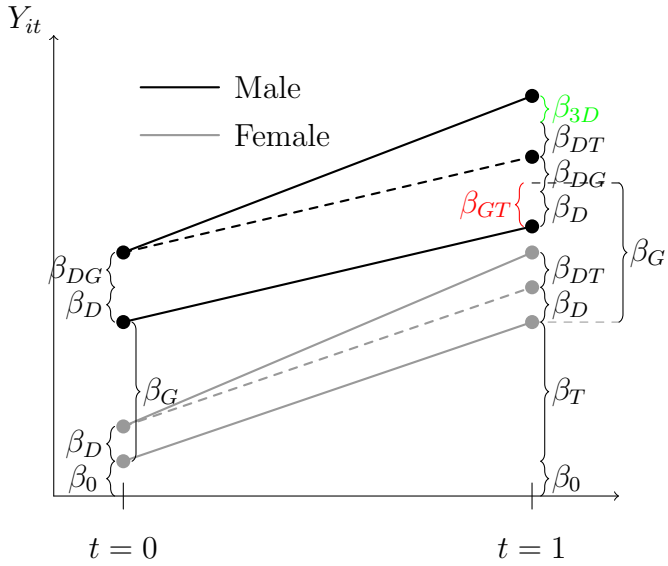
To grasp some intuition on the interpretation, consider evaluating an extended paternity leave policy. Suppose the policy affects *parents* more than *non-parents* (dimension G_i), but at the same time a tax reform occurs that affects treated and control locations differently over time. A DDD design compares: (i) treated vs. control locations, (ii) pre vs. post periods, (iii) parents vs. non-parents. Under appropriate assumptions, the third differencing removes confounding shocks that are common across the G_i dimension.

Figure 2 provides graphical intuition for the triple-differences regression in (7). The figure overlays two separate DiD comparisons, one for each subgroup G_i (illustrated as "Female" in gray and "Male" in black), and then tracks how the regression coefficients build up the mean outcomes cell by cell.

Start from the baseline group (female controls at $t = 0$): its mean is β_0. Moving to $t = 1$ within the same group adds the common post-period effect β_T. Switching from controls to treated within the female subgroup at $t = 0$ adds the treated-group level difference β_D, and in the post period it adds both β_D and the standard DiD interaction β_{DT}, so that the treatment effect for females is captured by β_{DT}.

The coefficient β_G captures the baseline level difference between males and females, common to treated and control and constant over time; this is shown by the vertical distance between the male and female control paths. The interaction β_{DG} allows the treated–control level difference β_D to differ across subgroups (males versus females). Finally, β_{GT} captures subgroup-specific time shocks: the

Figure 2. Triple differences: differencing out time shocks, treated-group differences, and group-specific differences.



differential change over time for males relative to females *even in the absence of treatment*. This term is precisely what can break a standard DiD design, and is differenced out by the triple-differences strategy.

The triple interaction β_{3D} is the parameter of interest: it measures the additional post-treatment change for treated units in the subgroup more exposed to the policy, *net* of (i) common time effects (β_T), (ii) treated-group differences (β_D), (iii) subgroup differences (β_G), and (iv) subgroup-specific time shocks (β_{GT}). Equivalently, β_{3D} can be viewed as the difference between two DiD effects (one within each subgroup), and is identified under the assumption that any deviation from parallel trends between treated and control groups is the same across the two subgroups.

V. Synthetic Control Methods

Synthetic control methods are particularly useful in settings with one (or a few) treated units and many potential controls, and with multiple pre-treatment periods. The key idea is to construct a "synthetic" control unit that reproduces the treated unit's *pre-treatment trajectory* as closely as possible.

Let $\mathcal{C}$ index untreated donor units. A synthetic control outcome is a weighted average:

$$Y_t^{SC} = \sum_{j \in \mathcal{C}} w_j \widehat{Y}_{jt}, \qquad w_j \geq 0, \quad \sum_{j \in \mathcal{C}} w_j = 1. \tag{8}$$

Let X_1 be a vector of pre-treatment predictors for the treated unit (often including lagged outcomes), and let X_0 collect the same predictors for donor units. Weights are chosen to minimize a weighted distance:

$$\hat{W} = \arg\min_{W} \left(X_1 - X_0 W\right)' V \left(X_1 - X_0 W\right) \quad \text{s.t. } w_j \geq 0, \sum_j w_j = 1. \tag{9}$$

The credibility of synthetic control relies on *excellent pre-treatment fit*: if the synthetic control tracks the treated unit closely prior to treatment, it is more plausible that it would continue to do so absent treatment. Inference often relies on placebo/permutation procedures (reassigning treatment to donor units) rather than standard asymptotics.

Classical synthetic control restricts weights to be non-negative and sum to one, limiting extrapolation but potentially yielding poor pre-fit if the treated unit lies outside the convex hull of controls. Modern extensions include:

- *Allowing negative weights:* can improve pre-fit but introduces extrapolation.
- *Augmented synthetic control:* combines synthetic control with bias correction when pre-fit is imperfect (often allowing negative weights).
- *Generalized synthetic control:* accommodates multiple treated units, multiple periods, and staggered adoption under latent factor structures.
- *Matrix completion methods:* treat counterfactual outcomes as missing data and recover them using machine-learning tools.

VI. Fixed Effects, Panel Data, and Two-Way Fixed Effects

There is *no need to have panel data* to estimate DiD. However, with panel data it is common to estimate a two-way fixed effects (TWFE) model:

$$Y_{it} = \beta_0 + \eta_i + \delta_t + \beta D_{it} + U_{it}, \tag{10}$$

where D_{it} is treatment status at time t, η_i is an individual fixed effect, and δ_t is a time fixed effect.

The term $\beta_0 + \eta_i$ captures permanent (time-invariant) differences across units, i.e. an individual-specific level of untreated potential outcomes. The time effects δ_t capture common aggregate time shocks (the average trend absent treatment). Under appropriate conditions, β identifies the ATT when treatment effects are constant and identification comes from within-unit changes in D_{it} over time.

When treatment adoption is staggered across groups (different units become treated at different dates), TWFE regressions such as (10) combine information

from many implicit comparisons. Some units act as controls at some dates and treated at others.

Goodman and Bacon (2021) show that, in staggered adoption settings, the TWFE coefficient can be written as a *weighted average* of all possible 2×2 DiD comparisons across cohorts and time windows. The weights depend on: (i) the sizes of the treated/control subsamples involved in each comparison, and (ii) the timing of treatment (comparisons nearer the "middle" of the sample window often receive more weight).

A key implication is that, with *heterogeneous treatment effects* over event time or across cohorts, the TWFE estimate can place *negative weights* on some comparisons. As a result, the TWFE coefficient can be difficult to interpret and can even flip sign relative to underlying causal effects. Moreover, changing the spacing/timing of adoption can alter the weights and therefore the overall estimate. If treatment effects are *constant across cohorts and over time*, these issues disappear and TWFE recovers the ATT.

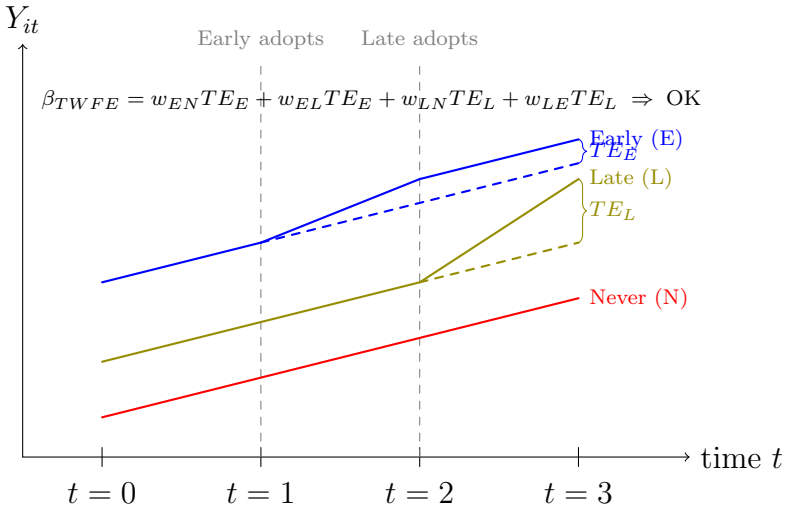
Figure 3 illustrates the Goodman–Bacon insight for staggered adoption designs estimated by two-way fixed effects (TWFE). With multiple adoption dates, TWFE combines several implicit 2×2 DiD comparisons: early-treated versus never-treated (EN), late-treated versus never-treated (LN), early-treated versus late-treated (EL, before late adoption), and late-treated versus early-treated (LE, after early adoption).

Panel (a) shows the *"OK"* case in which treatment effects are constant (or, more generally, do not vary with event time in a way that creates conflicting comparisons). In that situation, all 2×2 DiD components target the same causal parameter, so TWFE can be interpreted as a weighted average of sensible DiD estimands.

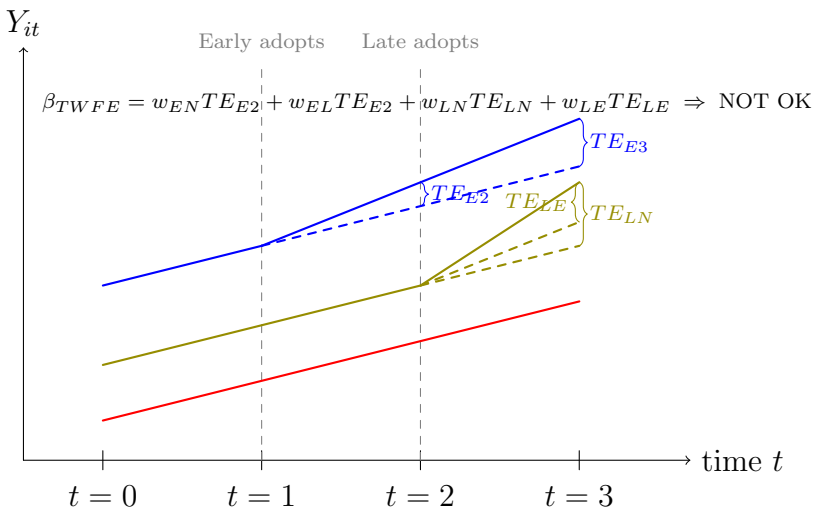
Panel (b) shows the *"NOT OK"* case with dynamic or heterogeneous treatment effects. Once the early-treated group has been treated for some time, it can still enter TWFE as a control for the late-treated group in post periods (the LE comparison). Because early-treated units are not valid controls for untreated outcomes of late-treated units when effects evolve over event time, TWFE may attach non-transparent (and potentially negative) weights to comparisons that do not have a clean causal interpretation. This is why, with staggered adoption, one must be careful interpreting TWFE estimates and often prefer estimators designed for heterogeneous and dynamic effects.

Figure 3. Goodman–Bacon intuition in staggered adoption designs.

(a) Constant treatment effects across event time: TWFE aggregates sensible comparisons.



(b) Dynamic/heterogeneous effects: TWFE mixes comparisons (including treated-as-controls), potentially with negative weights.



VII. Event Studies

A natural way to study dynamics and diagnose parallel trends is an *event study*. Let T_{0i} denote the period in which unit i first becomes treated. Define *event time*:

$$k \equiv t - T_{0i},$$

so that $k < 0$ are pre-treatment periods and $k \geq 0$ are post-treatment periods.

A standard event-study specification is:

$$Y_{it} = \eta_i + \delta_t + \sum_{k \neq -1} \beta_k \; \mathbb{1}\{t - T_{0i} = k\} + U_{it}, \tag{11}$$

where we normalize $\beta_{-1} = 0$ (the period just before treatment is the omitted category).

The interpretation is as follows:

- Each β_k measures the average effect k periods relative to treatment.
- Coefficients for $k < 0$ provide a *diagnostic for parallel trends*: they should be statistically close to zero if treated and control units follow similar counterfactual dynamics before treatment.
- Coefficients for $k \geq 0$ trace out *dynamic treatment effects*: timing, buildup, and persistence.

Event studies do not *prove* parallel trends, but they provide a transparent and informative diagnostic. Identification requires that, in the absence of treatment, outcome changes for treated and control units would be comparable in the pre-treatment event-time periods.

VIII. Shift–Share Instruments

Shift–share (or *Bartik*) instruments exploit variation coming from: (i) *shares*—predetermined exposure of units to sectors or groups—and (ii) *shifts*—aggregate shocks affecting those sectors or groups.

A canonical shift–share instrument has the form:

$$Z_{it} = \sum_k s_{ik} \, g_{kt},$$

where s_{ik} is unit i's exposure to sector k and g_{kt} is a sector-level shock common to all units.

Classic examples include predicted local labor demand shocks from national industry growth and predicted immigration inflows using historical settlement patterns. Identification relies on the idea that units differ in exposure to shocks that are common across locations, rather than on idiosyncratic local shocks that may be endogenous.

Shift–share IVs are *not automatically valid* just because shocks are aggregate. Common concerns include:

- shares s_{ik} correlated with unobserved local trends,

- shocks g_{kt} affecting outcomes through channels other than the endogenous regressor,
- concentration in a few sectors, so that shocks may themselves reflect local conditions.

Modern work clarifies that identification can be justified through (at least) two distinct routes:

- *Identification from exogenous shifts:* shocks g_{kt} are as-good-as-random, and identification comes from variation at the shock level (typically requiring many independent shocks). No strong exogeneity assumption is needed on the shares beyond being predetermined.
- *Identification from exogenous shares:* shares s_{ik} are valid instruments, while shocks g_{kt} primarily determine the (time-varying) weights placed on each share. Identification comes from cross-unit variation.

These approaches are not equivalent and can imply different inference strategies.